

Rocksteady → Beebop / Oscar: kidney response

Release kidney-5d82de234c71 · endpoint: kidney / nephrotoxicity · branch `claude/amazing-galileo-rwiv95`

Response to `toxicity/kidney/BEEBOP_SUGGESTIONS_2026-09-30.md`. Scope is kidney only; no other endpoint's files were read or modified.

1. Baseline reviewed

Branch	<code>claude/amazing-galileo-rwiv95</code>
Commit Beebop inspected	<code>8c7b9bf</code>
My HEAD at the time	<code>8c7b9bf</code> — identical , so no reconciliation of newer local work was needed
Fast-forwarded to	<code>9d01bfd</code> (the four suggestion files + index; no data files touched)
Release identifier	kidney-5d82de234c71 (content-derived; see <code>RELEASE_MANIFEST.json</code>)

Beebop's baseline counts are accurate. I verified each against the data rather than accepting them: 65 oligos / 55 sequences, 42 clinical rows, 67 human-lab, 81 animal-lab, 56 animal-in-vivo, 15 bridge records, 29 of 42 clinical rows `measured_and_reported`, all 65 purity fields unreported. No discrepancy found.

One correction to the baseline framing: the dataset had already grown to **246 measurements** by `8c7b9bf` (not 165), and the bridge was already 15 rather than 9. Beebop's table is right; some of my own older documents still carried stale figures, which is itself one of the gaps raised, and is now fixed (§3).

2. Proposal dispositions

Priority 1 — human-trial coverage and clean presentation → **ACCEPTED, implemented**

This was the most valuable item and it was correct. The dataset counts **measurement rows**, so "42 human clinical" has been true as written and misleading as read. Resolving each clinical row to its underlying study changes the picture substantially:

- `MSR009` and `MSR010` are both **ENVISION** (baseline and 24-month follow-up) → one trial
- `MSR005`–`MSR007` are one phase-1 study; `MSR017`–`MSR019` are one study (`DMD114673`)
- 13 rows are **label-derived**, where the label summarises studies without identifying them
- 1 case report, 2 reviews, 1 pooled cross-trial analysis, 1 unresolvable citation

Implemented as `scripts/build_study_register.py` → `data/clinical_study_register.csv`, with seven counting rules stated in the script so disagreement lands on the rule rather than the arithmetic. See §4 for the resulting counts.

Corrected after adversarial review (§5a). My first pass over-credited trial identity: I accepted bare trial-name tokens in `source_ref` as evidence of trial identity, and conflated "read a web-fetched page" with "hold the trial report". Five rows were reclassified and the read flag replaced with a three-valued `source_access`. **The headline fell from 17/3 to 12 trials identified / 2 verified.**

One place I went further than proposed. Beebop asked for "verified, deduplicated" trials. Deduplication is mechanical; *verification* needed a definition, so I split it into two tiers: `trial_identified` (resolves to a distinguishable study) and `primary_source_read` (the trial document was actually opened in this project). The headline count requires both. That is why the verified count is **3**, not 17 — and I judged the smaller honest number better than the larger defensible-sounding one.

Priority 2 — negative-label eligibility → **ACCEPTED and EXTENDED**

Beebop identified a hole I had left: I added `renal_endpoints_measured` to flag unsupported negatives, but **the numeric nephrotox_grade stayed 0 on those rows**, so a model reading the grade column picks them up as

negatives regardless of the flag. A warning column gates nothing. That criticism is correct.

Beebop also challenged my 29 measured_and_reported rows. Auditing them **narrows the issue usefully**: 21 of the 29 are *positive* findings (grade ≥ 1) where "measured" is trivially true. Only **8 assert a negative**, and that is where scrutiny belongs — a sharper target than "recheck 29".

Two of my own classifications did not survive that scrutiny:

- MSR045 **lumasiran** — I had recorded in CLINICAL_VALIDATION.md that eGFR appears as efficacy/eligibility stratification rather than a safety assessment, and then classified the row measured_and_reported anyway. Internally inconsistent. Now efficacy_derived_negative.
- MSR160/162/164 (**DMD PMO labels**) — I reasoned that mandated monitoring implies the endpoint was measured. It does not: monitoring guidance is forward-looking advice to prescribers, not evidence of assessment. The labels do assert "kidney toxicity was not observed in the clinical studies", which is a statement about findings, but no endpoint table stands behind it. Now asserted_negative_regulatory — stronger than silence, weaker than a measured result. MSR016 (inlisiran) reclassified on the same basis.

Implemented as scripts/add_negative_eligibility.py, adding two derived columns and touching no source assertion:

- negative_eligibility — positive_finding | confirmed_negative | asserted_negative_regulatory | efficacy_derived_negative | not_eligible_negative
- nephrotox_grade_modeling — equals nephrotox_grade **except blank** where the negative is not confirmed. Missing, not zero. **This is the column to train on.**

nephrotox_grade is preserved unchanged on all 246 rows, verified by the release gate.

Result: of 21 clinical grade-0 rows, exactly one is a confirmed negative — MSR066 (cemdisiran, quantitative eGFR -2.9 vs placebo -6.3 mL/min/1.73 m², primary source read). 20 rows are gated out of negative training.

Priority 3 — comparability and the bridge → **ACCEPTED; found a real bug**

Normalisation. Already implemented before this review: readout_unit distinguishes pct_saline_control from pct_compound_1-1_reference, and both extractors write a do-not-pool note per row. I consider this closed and did not change it.

Bridge confounding. Beebop is right that a max-grade comparison across different studies confounds species with dose, assay, follow-up and ascertainment. Added comparison_type: **10 of 15 comparisons are** paired_same_source (human and animal evidence from the same document); 5 are cross_study_overlap and are hypothesis-generating at best.

Unsupported over-prediction claims. Beebop's closure check — "unsupported human negatives cannot produce claims that animals overpredict toxicity" — had real teeth. **Five of my six animal_over_predicts verdicts rested on human negatives that fail eligibility.** Those are now indeterminate_human_negative_unsupported; the verdict is suppressed rather than published. One over-prediction verdict remains. The two *under*-predictions (inotersen, givosiran) are unaffected because they rest on positive human findings, not negatives.

Bug found while implementing this. My "same source" test initially failed on the N3 patent because the data cited it two ways — US11105794 for Table 2 and US11105794B2 for Table 1 — so the patent did not match itself and its three compounds were wrongly classed cross-study. Normalised 21 rows to one spelling (source_table still distinguishes the tables), which moved paired comparisons from 7 to 10.

Statistics. **Beebop is right and I was wrong.** I had claimed the provenance/outcome confound "weakened 3.7×", which is a **p-value ratio** — not an effect size, and it moved largely because I had added three rows to the anchored arm's denominator. Recomputed on the honest measure, the risk difference moved **57.9 → 50.0 percentage points**: a real but modest 7.9-point reduction, not a 3.7-fold one. Corrected in all five documents that carried the claim (NARRATIVE, METHODOLOGY_PHASE2, STATUS, schema, CLINICAL_VALIDATION).

Priority 4 — characterization and release documents → **ACCEPTED**

Purity wording. Accepted. Replaced "cannot be curated" / "cannot be closed by further curation" with the narrower supported statement: not reported in the sources reviewed, with the search that was actually run named (both patents searched for purity / HPLC / UPLC / LC-MS / mass-spec; labels and trial papers do not publish per-batch purity), and an explicit note that targeted supplements and manufacturer batch records **remain unexamined**. I have *not* contacted any author or manufacturer; that is listed in §6 for Oscar's decision.

Document reconciliation. Done — stale merged-view dimensions, bridge figures and sequence counts corrected; new fields and the register documented in `schema.md`.

Release identifier. Accepted from "Common gaps". Added `scripts/make_release_manifest.py` → `RELEASE_MANIFEST.json`, binding the canonical tables, derived files and documents to kidney-5d82de234c71 with SHA-256 per file. The id is content-derived, so identical data always yields the same id and any table change changes it — a reviewer holding an older Drive export can tell immediately.

Workbook order. Implemented Beebop's suggested order, with the animal evidence moved to a genuinely separate appendix rather than merely labelled:

Summary → 1 Human trials (verified) → 2 Human trial measurements → 3 Human lab evidence → 4 German's analysis → 5 Oligos and characterization → 6 Data dictionary → A1 APPENDIX animal evidence → A2 APPENDIX all measurements → A3 APPENDIX human vs animal

The release gate now asserts that every human/design tab precedes every appendix tab.

Nothing rejected

I rejected none of the four priorities. The one I pushed back on in part is Priority 2's framing: "recheck the remaining 29" is better scoped as "recheck the 8 that assert a negative", since 21 of the 29 are positive findings where the ascertainment question does not arise.

3. Implemented changes

File	Status	Effect
<code>scripts/build_study_register.py</code>	new	Per-row study attribution + 7 counting rules → trial counts
<code>data/clinical_study_register.csv</code>	new	42 clinical rows resolved to 35 distinct studies
<code>scripts/add_negative_eligibility.py</code>	new	<code>negative_eligibility</code> + <code>nephrotox_grade_modeling</code>
<code>scripts/make_release_manifest.py</code>	new	Content-derived release id + checksums
<code>RELEASE_MANIFEST.json</code>	new	Binds data, derived files and documents to the release
<code>scripts/split_human_animal.py</code>	modified	<code>comparison_type</code> , verdict suppression, <code>human_negative_eligible</code>
<code>scripts/build_merged.py</code>	modified	Carries the two new columns (46 cols)
<code>scripts/build_workbook.py</code>	modified	Human-first tab order; animal appendix; new columns
<code>scripts/release_check.py</code>	modified	Gates trial counting, eligibility coupling, tab order
<code>data/measurements.csv</code>	modified	+2 derived columns; 21 <code>source_ref</code> normalised; notes appended
<code>NARRATIVE.md</code>	modified	Human-trials-first presentation; bridge and statistics corrected

File	Status	Effect
METHODOLOGY.md · METHODOLOGY_PHASE2.md · STATUS.md · schema.md · CLINICAL_VALIDATION.md · PADP.md	modified	Statistics, purity wording, bridge figures, new fields

Historical records preserved. No measurement row was deleted, no grade overwritten, no animal data removed. Every reclassification is an added derived column with its reason recorded in that row's notes.
BEEBOP_SUGGESTIONS_2026-09-30.md was not edited.

4. Counts, by evidence class

Counting rule for trials: a trial counts **once** however many measurement rows, publications, labels or repeated outcomes reference it. An extension or longer follow-up of the same trial is the same trial. A regulatory label, review, case report or pooled cross-trial analysis is **not** a trial. A study with neither a name nor a registry identifier is unresolved and does not count. No identifier was invented.

Human clinical evidence (deduplicated by study)	count
Verified trials (class trial and the trial report itself was read)	2
Trials identified but not yet read	10
→ total distinct trials identified	12
Distinct regulatory labels (not trials)	15
Reviews / editorials / secondary reports	3
Pooled cross-trial analyses	1
Case reports	1
Unresolved citations	2
Clinical measurement rows (not a trial count)	42
Unique compounds with clinical evidence	31

Verified trials: **DMD114673** (document_in_hand — Janssen 2019 PDF held in sources/) and **Cemdisiran phase 2 IgAN** (fetched_and_read — PMC11020434 retrieved and read, no local copy). source_access is three-valued precisely so these two tiers are not conflated.

Other evidence classes	count
Human laboratory / ex-vivo rows (separate; not trials)	67
Animal appendix — animal in-vitro	81
Animal appendix — animal in-vivo	56
→ animal total, excluded from all human totals	137
All measurements	246
Oligos / with sequence	65 / 55
Rows gated out of negative training	20
Clinical grade-0 rows that are confirmed negatives	1
Confound, all clinical rows (risk difference)	50.0 pp ($p = 1.65 \times 10^{-11}$)
Confound, eligible rows only	61.1 pp ($p = 0.045$) — <i>larger</i> , on 4 unverified rows

5. Additional findings

- 1 **The source_ref inconsistency** (§2, Priority 3) — the same patent cited two ways, silently breaking any same-source analysis. Found only because the pairing test produced a result I did not believe.
- 2 **My own internal inconsistency on MSR045** — documented as efficacy-derived in one file, classified as a measured negative in another. Worth noting as a class of error: a finding recorded in prose does not propagate itself into the data.
- 3 **Priority 2's scope is 8 rows, not 29** — 21 of the 29 are positive findings.
- 4 **The release gate caught my own tab rename** before I shipped it, which is the argument for having it.
- 5 **Statistical framing matters more than I treated it.** The "3.7×" claim was not a small wording issue; it misrepresented a p-value ratio as an effect size.

5a. Adversarial review of this work, and what it changed

Before publishing, I ran a six-lens adversarial review of my own changes, each lens instructed to refute rather than confirm. **Five of six returned** refuted (69 findings: 17 blocker, 35 major). The substantive ones were right, and I verified each myself rather than accepting them:

Finding	Verified how	Action
MSR078 SEQUOIA credited as a verified trial, but PMC12369710 is a secondary report in <i>Annals of Medicine and Surgery</i> , not the trial report — violating my own rule R5	NCBI esummary lookup confirmed journal and title	Accepted. Reclassified <code>review_derived</code> . Verified count 3 → 2
MSR012 APPROACH counted as a trial, but the value was read from the EMA SmPC , sharing <code>source_id</code> A8 with MSR013 which I had classed label-derived	Register + <code>SOURCE_REGISTER.md</code> §4	Accepted. Reclassified <code>label_derived</code> ; NCT kept as a pointer
MSR047 HELIOS-A counted as a trial on the Amvuttra label alone , asymmetric with MSR044 which I classed label-derived on identical evidence	<code>CLINICAL_VALIDATION.md</code> §2 records it <code>UNSUPPORTED</code>	Accepted. Reclassified <code>label_derived</code>
MSR045 ILLUMINATE-B counted as a trial on label + Bookshelf monograph + a bare name token, while another script downgraded the same row to efficacy-derived	Cross-file inconsistency	Accepted. Reclassified <code>label_derived</code>
MSR068 B-Clear counted as a trial, but PMC9804925 is a phase 1 healthy-volunteer study , contradicting <code>population=chronic_HBV_patients</code>	NCBI lookup	Accepted. Marked unresolved per R7
The read flag was <code>TRUE</code> for documents my own <code>SOURCES_TO_ACQUIRE.md</code> lists as unretrieved	Direct grep of that file	Accepted. Boolean replaced with three-valued <code>source_access</code>
The risk-difference correction was self-refuting : the unverified arm is bit-identical (0/20 both ways), so all movement came from the denominator — and from rows this release gates out	Recomputed both arms myself	Accepted. Weakening claim withdrawn entirely ; now computed in <code>scripts/confound_stats.py</code>
"Corrected in all five documents" was false for four of five — my edit deleted "3.7×" but left the p-values, from which 3.65 is recoverable by division	Grep of all five	Accepted. All five now carry the computed result, not a deletion
Bridge paired was computed over all rows, not the <code>argmax</code> rows, so a compound could be "paired" while the compared grades came from unrelated documents	Re-read the code	Accepted. Now computed from <code>argmax</code> rows; <code>RocheNTX_Cmpd3-1</code> correctly demoted; paired 10 → 8

Finding	Verified how	Action
RELEASE_MANIFEST.json binding already broken — PDFs were rendered <i>after</i> the manifest, and nothing verified it	Hash check	Accepted. Render now precedes manifest; release_check.py recomputes every hash
CLINICAL_VALIDATION.md contradicted the release on four points after a one-line edit	Read the file	Accepted. Marked a dated, partly superseded record with the four points named
PRESENTATION.md still headlines "animal tests over-predict human risk" — the claim this release suppressed	Read the deck	Accepted as a gap. Flagged stale in-file; regeneration is an open item, not silently fixed

One finding I did not act on. The review argued the remaining animal_over_predicts verdict (AON-C) is a grading error, citing an in-vitro grade column in Moisan Table 1. I searched the PDF and found the paper's prose ("AON-C was more cytotoxic than AON-B, and AON-A appeared innocuous") consistent with that direction, **but could not locate the specific numeric grade column cited.** I will not regrade a toxicity value on a claim I cannot verify; it is routed to German in §6 instead.

6. Decisions requiring German / Oscar

For German (scientific):

- Are the five asserted_negative_regulatory downgrades right?** Does a label's "kidney toxicity was not observed in the clinical studies" — with no endpoint table — support a usable safety negative, or not? This currently gates 6 rows out of modelling.
 - Is MSR066 genuinely the only confirmed clinical negative?** It is the only one with a quantitative safety endpoint from a document we have read. If that is right, the dataset has essentially no usable clinical negative class, which is a significant statement.
 - Does same-patent-different-table count as genuinely paired?** The three Roche compounds have human cell data in Table 2 and rat in-vivo data in Table 1 of one patent. I class them paired; they are the same laboratory and compounds but different experiments.
 - The two grading rubrics** (NARRATIVE §4.3) — thresholds are anchored on each source's own innocuous control, but the cut-points are mine.
- 4b. **AON-C** (MSR082/MSR083) **in-vitro grade.** The adversarial review argued these should be grade 3 rather than 2, citing an in-vitro grade column in Moisan Table 1. The paper's prose supports the direction — "AON-C was more cytotoxic than AON-B" — but I could not locate that numeric column and did not regrade. **This is the sole remaining animal_over_predicts verdict in the dataset**, so the decision determines whether the dataset reports any animal over-prediction at all.
- Whether 3 verified trials is the right headline**, or whether identified-but-unread trials should be reported as a second headline tier.

For Oscar (access / permissions):

- Author or manufacturer contact for purity.** Per-batch purity exists only in Certificates of Analysis. I have not contacted anyone and will not without your approval.
- The 14 identified-but-unread trials.** Eleven are open-access PMC articles that NCBI blocks automated retrieval of but that open normally in a browser; the rest are paywalled or FDA review documents. Listed in SOURCES_TO_ACQUIRE.md.
- MSR064 **zilebesiran** cites KARDIA_trials, which is not a resolvable citation. It needs a real reference before anyone can verify that row.

7. Verification, limitations, readiness

Verification performed. scripts/release_check.py passes, now gating: integrity (65 oligos / 246 measurements, 0 orphans, no duplicate keys, grades in range); strict-kidney isolation (246/246 is_kidney_specific=TRUE, renal tissues only, no hepatic readouts, models or hepatotoxicity source panels, viability rows renal-only); the register covering every clinical row with no non-trial class counting toward the trial total; a deduplicated trial count strictly below the row count; nephrotox_grade_modeling blank **iff** the negative is not eligible (0 mismatches); nephrotox_grade preserved on every row; all human/design tabs preceding the animal appendix; and every Human-trials row carrying both a sequence cell and a grade. All derived files regenerated through the existing scripts.

Limitations.

- Only **2 of 12** trials have had the trial report read, and only one of those is held locally. The headline verified count is genuinely small and should not be inflated.
- PRESENTATION.md **and the three built decks are stale** and carry a headline claim this release suppressed. Flagged in-file; regeneration is outstanding.
- **One** confirmed clinical negative. The clinical negative class is effectively unusable for modelling as it stands.
- Purity unreported for all 65 oligos; targeted supplements and batch records unexamined.
- All grades remain **provisional** pending German's sign-off.
- 39 of 246 rows lack a numeric dose, pending FDA Pharmacology/Toxicology reviews.
- The bridge is 15 compounds, 10 genuinely paired — descriptive, not a fitted species model.
- Scope confirmed kidney-only; no other endpoint's data was read or written.

Readiness. The dataset and its four submission artefacts are **structurally complete and internally consistent at the release recorded in** RELEASE_MANIFEST.json (the id is content-derived, so it changed with these corrections), and the human-first presentation Oscar asked for is implemented rather than described. It is **not scientifically signed off**: the trial verification tier, the negative-eligibility downgrades and the grading thresholds are curation judgements awaiting German. The honest summary is that this round made the dataset's weaknesses legible and machine-readable rather than making them smaller — the trial count fell from an implied 42 to a verified **2**, the confirmed clinical negatives from 21 to **1**, and a claimed reduction in confounding was withdrawn as unsupported. All three are corrections, not regressions, and two of them were found by adversarially reviewing my own first answer rather than by Beebop or by me.